

# Research Project: COCONUT

## Roomify, A Co-Creative Interior Design Tool

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### Abstract

It can be difficult to imagine what a room would look like with your own furniture in it. To help solve this problem, a tool was developed which allows users to upload floorplans of apartments, place furniture inside of the virtual rooms, and generate 3D visualizations with the help of AI. This co-creating process was intended to inspire the user with new ideas and possibilities they might not have thought of before. Testing and evaluations have shown that the AI, while prone to making mistakes and showing a lack of understanding of what the user intended, sometimes leading to frustration, did inspire users with its choices and made the user feel in charge. This challenges the assumption that a co-creative AI must be smart or perfectly compliant to be effective.

### Introduction

When looking for a new home, it might be difficult to visualize what the rooms would actually look like with your own furniture in them. Often, the only pictures available to users are ones that contain pre-furnished rooms, making it hard for users to imagine how their furniture would fit in. Blank floorplans, while helpful, are also not sufficient.

Using AI technology, we propose and showcase a tool which allows users to upload an .fml file and work together with a computer agent to sketch out a layout of furniture and create a realistic, 3D visualization in a specific design style. The computer agent helps the user with not only the creation of the visualization, but also by suggesting possible furniture placements, leading to a back-and-forth co-creating cycle between human user and computer agent. In addition to accelerating the workflow of both amateur and professional interior designers, the tool will also hopefully lead to inspiration of new ideas as a byproduct of the computer agent's suggestions.

### Research Process

We researched three areas we deemed important to the formation of our eventual co-creating tool. These are: (1)

the usage of AI in home interior design tools, (2) common UX/UI practices in home interior design tools, and (3) the differences between interior design styles around the world. Many aspects of this research have played an influential role in the eventual design decisions regarding the creation of our tool, which will be described in detail in section 3 of this paper.

### AI and Interior Design Tools

Many different interior design tools for both professional and layman users have incorporated AI technology to a certain extent. Tools such as Planner5D, Spacely AI, and Foyr Neo all boast about their ability to quickly generate 3D models and realistic renders of user inputted 2D floorplans, allowing for fast concept designs. Regarding furniture, Planner 5D is capable of using AI to automatically place furniture based on the geometry of the room. This usage of AI to arrange furniture has been researched before in 2019 (Yang and Li), where certain areas of a floorplan were assigned categorized lists of furniture items (e.g., beds and nightstands for “sleeping area”). Other research focused more broadly on the topic of co-creating an interior design with AI has shown that users reported increased satisfaction and creativity when collaborating with an AI agent (Jiang, 2025).

Other tools such as RoomGPT are focused more on simply helping users visualize what a room would look like with different furniture, without having the capability to create and edit a floorplan. This functionality is often called virtual staging and, in the case of RoomGPT, is catered to a more casual, non-professional audience. Virtual staging can also be found in the above-mentioned tools.

### UX/UI Design

#### Furniture Selection and Placement

Shneiderman et al. (2016) have shown evidence that categorized, searchable libraries of items improve the workflow efficiency of non-experts and experts alike. Other sources support this claim by asserting that a centralized

catalog also increases the reliability of the tool (UXPlanet, 2023). Introducing a (grid-like) system where the items can be manipulated in space, such as by dragging and dropping them, further reinforces user performance (Hutchins et al., 1985).

### Viewing Methods

The manner in which a user is capable of viewing the virtual layout they are creating has been shown to have a large impact on their usage of the tool. A real-time 2D view of whatever a user is manipulating, for example, allows for quick error correction, improving precision (Nielsen, 1994). Combining this with other viewing methods such as 3D, AR, and VR further increases the spatial awareness of the user (Azarby and Rice, 2022). However, eye-tracking experiments have found that, in design systems with both a 2D and a 3D view, users spent most of their time (80%) focusing on the 2D visualization (Rusnak, 2022). Further research shows that 3D visualizations of a scene, while increasing awareness, also introduce more cognitive load in the user (Che et al., 2025).

### Interior Design Styles

There exist many different interior design styles across the world, all being influenced by various parts of their country of origin's culture. These influences can be found in not only the appearance of the furniture items (e.g., color, size, complexity of design), but also in their placement and abundance. Some cultures' interior design styles even use completely different types of furniture compared to others, such as the floor sofas used in Middle Eastern countries. However, while there can be a massive difference in design philosophy, common ground can also be found between many styles. The Japanese and Scandinavian interior design styles, for example, are commonly known to emphasize minimalism, keeping the amount of furniture within a room to a certain degree. In fact, these similarities have given birth to the interior and architectural design style of Japandi, which incorporates both the Japanese and Scandinavian design philosophies.

Regarding the placement of furniture in particular, the Chinese philosophy of feng shui contains a set of interior design rules that are commonly upheld across many different countries, even making its way over to the west in modern times. Feng shui is focused on improving the "flow of energy" of a space (in this case, a room) and bettering the interaction between human and environment. Prominent guidelines include not placing a bed directly across from a door and emphasizing symmetry.

### Product Description

We created a web-based interior design support tool called Roomify utilizing a collaborative and human-in-the-loop approach for our research study findings. We used React for the frontend UI and Node.js for the backend which communicates with OpenAI LLMs to suggest logical layouts. This allows for a top-down, planimetric view for all

editing actions. Users import a floor plan (.fml format) upon opening the tool, which is then parsed into real time room geometry, walls, and functional areas (living room, dining area, etc.).

The tool has a categorized asset library located in the sidebar containing all furniture categories (sofas, tables, lighting, etc.). For both ease of use and accuracy, the workspace uses a snap to grid functionality for dragging and dropping furniture. Users can manually place furniture items, rotate furniture items, and edit dimensions to be equivalent to real world furniture dimensions.

### Generative Capabilities

The primary innovation of the tool is its generative capability. Upon selecting the "generate" option, the tool captures the current 2D floor plan in visual form, sends it to the backend along with the geometric wall information, and the AI agent analyzes the visual boundaries of the room and available furniture items to create a logical layout.

Using its ability to interpret the visual boundaries of the room and the furniture items currently available to the user, the AI agent is able to generate a valid layout for the space. In addition to being able to generate layouts based on user input, the tool also has the capacity to support two different forms of co-creation. First, the "locking" mechanism provides a means for the user to express their intent to the AI. The user may "lock" a furniture item, effectively treating it as a constraint that cannot be changed. Second, the tool analyzes the number of each type of furniture that the user has placed in the room and uses that information when creating new layouts, so that the new layouts always have the same number of each type of furniture. Once the AI is activated, it treats the user's locked furniture items as obstacles, and the unlocked furniture items as furniture that needs to be arranged. Therefore, the AI is able to "fill in the gaps," and generate new furniture arrangements that are consistent with the user's previous placements and furniture arrangement preferences. This capability transforms the generation process from a one-way service to a co-creative dialogue process. For example, if the user places a television and locks it, the AI can then use the location of that unit to place a sofa and coffee table that would allow someone to watch TV from that unit.

The visualization and immersion capabilities of the system are provided through the ability to toggle between a 2-dimensional representation (the original planning interface) and a 3-dimensional representation of the space. Upon activation of the 3-dimensional mode, the 2-dimensional representation of the floor plan becomes a fully navigable 3-dimensional environment. This functionality allows the user to perform spatial verification without having to continually visualize the 3-dimensional environment in real time. The user can transition between a camera view and a first person view while in the 3-dimensional environment. When using a camera view, the user can select to render a realistic 3-dimensional view of the camera view that is consistent with the selected interior style.

To accomplish this, the user sends the screenshot of the camera view and the selected interior style to gemini-2.5-pro (via Google Generative AI SDK), which generates a detailed prompt. That prompt is then sent to xlabs-ai/flux-dev-controlnet (via Replicate), which generates a high-quality, realistic 3-dimensional render of the camera view in the specified style, such that the furniture is located in the same locations as in the 2-dimensional environment. We decided not to implement the grid snapping functionality because this resulted in a negative user experience during our own testing phase.

## Implementation

Our use of Google Gemini allowed it to be our primary source of code generation; therefore, we designated the lead developer as AI and ourselves as the project managers. The primary focus of our contributions was to define the "vibe" of the system and its functionality through natural language input, then implement the AI-recommended code structure with only a few minor manual edits. Although the AI produced a large amount of code during this time, the overall process of developing the system was difficult and lengthy. Due to the numerous iterations and refinements needed to collaborate, we found that there were many instances where the AI misinterpreted certain requirements or developed outdated syntax. These instances required that we perform additional iterative refinement, critical thinking, and prompt engineering to produce functional results.

## Evaluation and Results

We applied Ritchie's (2007) framework for computational creativity (Appendix I) to evaluate Roomify, focusing on

typicality, value, and novelty. Data were drawn from questionnaires (Figure 1), behavioral observations, and qualitative feedback provided by three participants (homeowners aged 25–59) during a structured co-creative design task with AI.

## Typicality

Regarding typicality, which is the extent to which artifacts are recognizable members of a target class, the AI consistently produced identifiable living room configurations. This meets Ritchie's Criteria 1–2. Participants treated these outputs as legitimate starting points. However, moderate usability ratings (3, 3, 2) and qualitative reports of "unrealistic" object placement indicate a limitation. While the system maintains typicality, it struggles with strict spatial constraints. Criterion 9 requires nuance, as the system uses an LLM. The training data serves as an implied set of valid layouts. While spatial recommendations generally aligned with design ideals, the AI failed to consistently replicate the logical constraints of this dataset. Occasional hallucinations and inaccurate furniture orientations indicate that, while the model captures aesthetic norms, it struggles to enforce structural logic.

## Value

In co-creative systems, value is defined by user agency as much as the utility of the artifact. Participants reported high *Ownership* scores (5, 4, 5), meeting Criteria 4–5 regarding the production of valued outputs. Crucially, users viewed the final designs as personally meaningful. *Collaboration* scores were more varied (2, 3, 4), reflecting individual differences in engagement. One participant rejected most suggestions, while another went back and forth with

## Ritchie's Empirical Criteria for Attributing Creativity

1. *On average*, the system should produce **suitably typical** output.
2. A *decent proportion* of the output should be **suitably typical**.
3. *On average*, the system should produce **highly valued** output.
4. A *decent proportion* of the output should be **highly valued**.
5. A *decent proportion* of the output should be both **suitably typical** and **highly valued**.
6. A *decent proportion* of the output is **suitably atypical** and **highly valued**.
7. A *decent proportion* of the **atypical** output is **highly valued**.
8. A *decent proportion* of the **valuable** output is **suitably atypical**.
9. The system can **replicate** many of the **example artefacts** that guided construction of the system (*the inspiring set*).
10. *Much of the output* of the system is **not in the inspiring set**, so is **novel** to the system.
11. **Novel output** of the system (i.e. *not in the inspiring set*) should be **suitably typical**.
12. **Novel output** of the system (i.e. *not in the inspiring set*) should be **highly valued**.
13. A *decent proportion* of the output should be **suitably typical** items that are **novel**.
14. A *decent proportion* of the output should be **highly valued** items that are **novel**.
15. A *decent proportion* of the **novel output** of the system should be **suitably typical**.
16. A *decent proportion* of the **novel output** of the system should be **highly valued**.
17. A *decent proportion* of the **novel output** of the system should be **suitably typical** and **highly valued**.
18. A *decent proportion* of the **novel output** of the system should be **suitably atypical** and **highly valued**.

Figure 1. Graeme Ritchie's criteria for attributing creativity.

realistic AI visualizations and the drawing board. This variation suggests the system supports Criteria 6–8. Value is often indirectly derived from the user's reflection on suggestions.

## Novelty

The system performed strongest in the dimension of novelty. High *Exploration* scores (4, 4, 5) confirm the AI proposed layouts participants admitted they would not have conceived on their own (Criterion 10). Notably, placements initially described as weird or annoying were often later cited as helpful for gaining new inspiration. This observation supports Criterion 18, where a subset of atypical outputs drives the creative process by disrupting habitual design patterns.

## Discussion

These results suggest the system functions effectively as a lateral thinking partner. The system is most usable for exploratory creativity rather than strict adherence to spatial constraints. The primary evidence of creativity here lies in the co-creative process itself. Novelty here serves to invite reflection rather than compel adoption.

A key finding is the divergence between low understanding of the user's intent by the system and high *Ownership* scores. This challenges the assumption that a co-creative AI must be smart or perfectly compliant to be effective. Even when the AI failed to fully grasp the user's ideas, it succeeded as a provocateur that prevented design fixation. The high ownership scores confirm that the system avoids the overshadowing effect common in generative tools. The human remains the lead designer and has the final say.

However, the tension between typicality and value requires attention. While weird suggestions aided novelty, a lack of spatial realism frustrated users with specific constraints. Future iterations must aim to enforce stricter physical rules without sanitizing the edginess that fuels novelty.

Overall, the evaluation indicates that Roomify was perceived as a helpful and engaging co-creative tool. Despite its limitations, participants responded positively to the system and were able to effectively integrate it into their design process.

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Appendix I

DEEL 1: Instructie & Vragenlijst (Voor de Deelnemer)

Test: AI Co-Creative Interior Design

Bedankt voor je deelname. Deze test duurt ongeveer 10-15 minuten.

A. Het Scenario

Lees onderstaande opdracht goed door voordat je begint:

De Opdracht:

Stel je voor: je hebt een nieuw appartement gekocht waarvan je de plattegrond bezit. De WOONKAMER moet worden ingericht en daar wil je een plan voor maken.

Jouw doel: Richt de WOONKAMER zo in dat er een duidelijke plek is om te eten én een plek om te relaxen.

Spelregels:

- 1. Je mag meubels slepen en plaatsen.
- 2. Je moet de "Generate" (AI) knop minstens 3 keer gebruiken om ideeën op te doen.
- 3. Probeer samen te werken met de AI om tot een ontwerp te komen waar jij blij mee bent.

(Na afronding van de taak, vul onderstaande vragen in)

B. Jouw Ervaring

Geef aan in hoeverre je het eens bent met de volgende stellingen op een schaal van 1 tot 5.

(1 = Strongly Disagree, 5 = Strongly Agree)

| Categorie     | Stelling                                                           | 1                     | 2                     | 3                     | 4                     | 5                     |
|---------------|--------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Exploration   | "The AI suggested layouts I wouldn't have thought of myself."      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Collaboration | "I felt like I was working with a partner, not just a tool."       | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Ownership     | "Even though the AI helped, I feel like the final design is MINE." | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Usability     | "The AI understood what I wanted to achieve."                      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

The AI influenced my design choices

C. Slotvraag

Wat was het meest verrassende (positief of negatief) dat de AI deed tijdens het ontwerpen?